Reuse-Aware Layer Skipping (RALS) Documentation

1、Project Objective

RALS is a profiling tool designed to analyze the reusability of Transformer attention computations. Its primary goal is to identify: 1) which layers can safely skip redundant computation (skip layers); and 2) which layers must be recomputed to preserve model correctness (rollback layers), thereby reducing overall inference cost.。

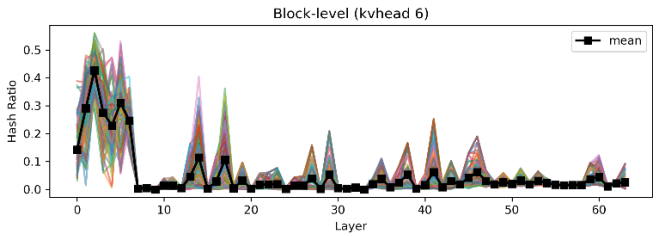
To achieve this, RALS analyzes：
• the sparsity of attention energy distributions, and
• the degree of attention overlap across different layers.

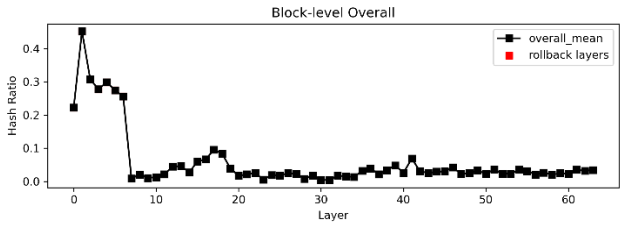
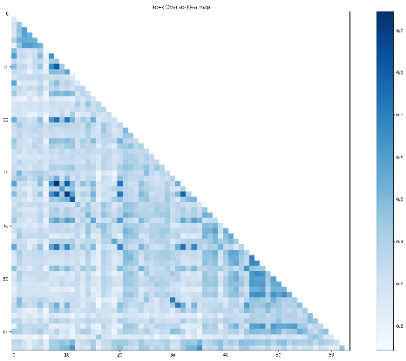
Based on these statistics, it automatically derives：
- rollback layers (layers that require full attention), and
- reuse paths for skip-layer execution（reuse path）

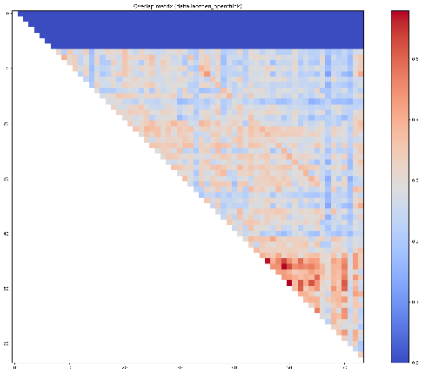
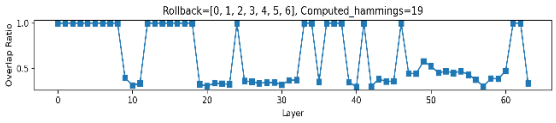
2、Overall Workflow

Load K/Q tensors
↓
Compute attention sparsity（hash ratio）
↓
Identify unstable layers（rollback layers）
↓
Compute inter-layer attention overlap matrices
↓
Search for the optimal reuse path
↓
Generate skip-layer execution configurations

3、Detailed Explanation of Generated Figures

hratio_{kvhead}.png (one per KV head)	 **Meaning:**

	Layer-wise attention sparsity profiles. **Figure components:** **X-axis:** Layer index (model depth) **Y-axis:** Hash Ratio — the proportion of attention blocks required to cover 90% of total attention energy **Light-colored curves:** distributions across individual decoding steps **Black bold curve:** temporal mean over all steps **Intuition of Hash Ratio:** **Small values (e.g., 0.1):** 90% of attention mass concentrates on a small number of tokens (high sparsity) **Large values (e.g., 0.8):** attention is widely spread, approaching global scanning **What can be observed?** ✔ Layers with highly concentrated attention (good candidates for reuse) ✔ Layers with large temporal variance (unstable behavior) ✔ Whether deeper layers tend to exhibit denser attention patterns **Purpose:** Provides statistical evidence for identifying rollback layers.
hratio_overall.png (aggregated across all heads)	 Meaning: Overall layer-wise attention sparsity after aggregating all KV heads. Used to determine layers with consistently low reusability that require recomputation.
top{K}overlap{head}.png (attention overlap heatmaps)	

	Meaning: Inter-layer similarity of attention focus patterns. **Each pixel represents:** the proportion of overlapping top-k attention blocks between Layer i and Layer j. **Purpose:** ✔ Identify layers with highly similar attention behaviors ✔ Enable direct reuse of KV results between those layers
matrix_{name}.png (merged overlap matrix)	 This figure merges overlap matrices across heads: **MLA mode:** mean aggregation **Standard mode:** minimum aggregation (conservative estimate) **Meaning:** The most reliable inter-layer attention reuse structure of the model. **Purpose:** Serves as the input graph for the subsequent optimal reuse path search. **Intuition:** It can be viewed as a “map” indicating which layers can safely skip computation by reusing results from earlier layers.
result_final_{idx}.png (final skip strategy visualization)	 **Meaning:** Quality of the selected attention reuse path. **X-axis:** Layer index **Y-axis:** Overlap ratio Where:

	1.0 indicates direct computation < **1.0** indicates reuse from previous layers **Purpose:** Provides an intuitive visualization of: ✔ Which layers are explicitly recomputed ✔ Which layers are skipped via reuse ✔ Whether reuse strength remains stable across the model

4、Interpretation of the Final Output Configuration

CONFIG 0:

[-1, -1, 1, 1, -1, 4, ...]

Configuration Rules：

Value	Meaning
-1	The layer is explicitly recomputed
k	The layer reuses the attention results from layer *k*